Question 1

a. HCI concept

Frequency of use. The designer placed each option according to how often it is used. Withdraw and Deposit are used constantly, so they get the main screen, while Change PIN and Print Statement go one level down. Cognitive load and menu design are results of this choice, not the reason for it, and task analysis is only how you would find the frequencies.

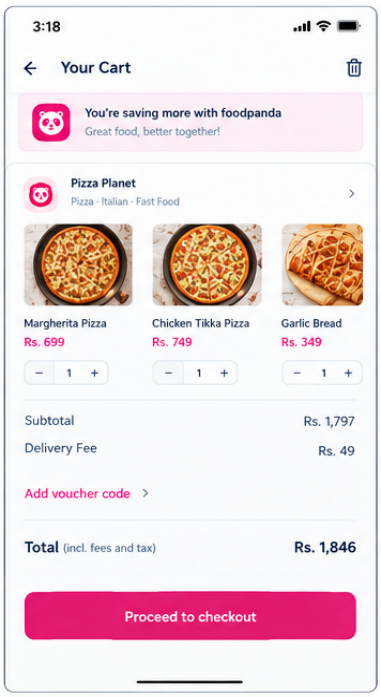
b. Design rationale

- Most visits are for cash, so the main task takes the fewest steps.
- A rare option can afford one extra tap. Nobody changes their PIN daily.
- A short list is easier to scan and keeps short term memory load low.
- Big buttons in obvious places are quicker to hit (Fitts' Law).

c. How it improves usability

Principle	**How the design supports it**
Learnability	A new user sees Withdraw and Deposit straight away, and they match what people expect from an ATM (familiarity), so no training is needed.
Flexibility	Frequent users get a short path to their main tasks, and the secondary menu still keeps other services available. The bank could even let users pick their own favourites (customizability).
Robustness	The main screen covers what people come to do (task conformance), a clean screen is easy to read (observability), and fewer steps feel quick (responsiveness).

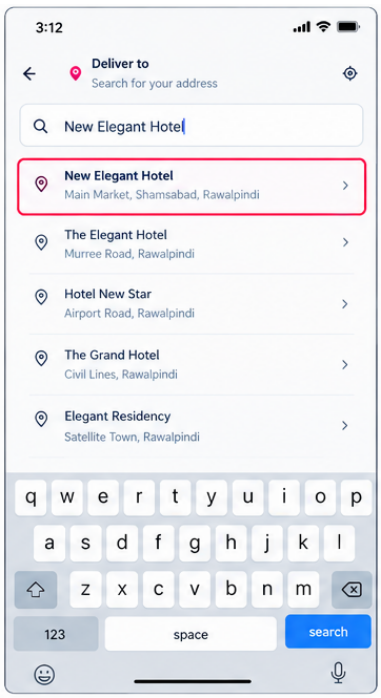
Question 2
a. Usability problems



1. Tiny quantity buttons

The plus and minus buttons in the cart are very small and close together, so they are hard to hit with a thumb.

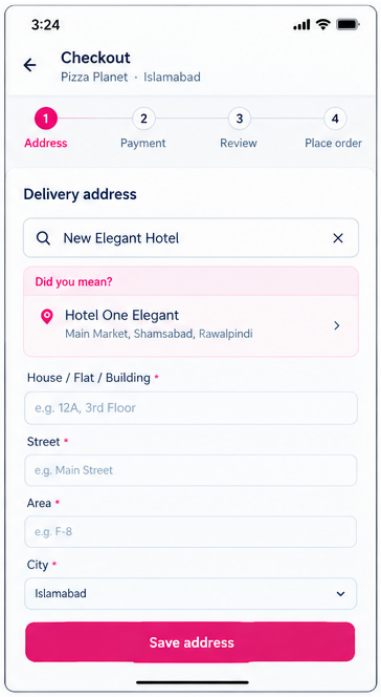
Principle violated: Fitts' Law (target size)



2 . Wrong address suggestion

The search lists several similar hotels, and the checkout screen then asks "Did you mean Hotel One Elegant?" for the same search. It is easy to pick the wrong address.

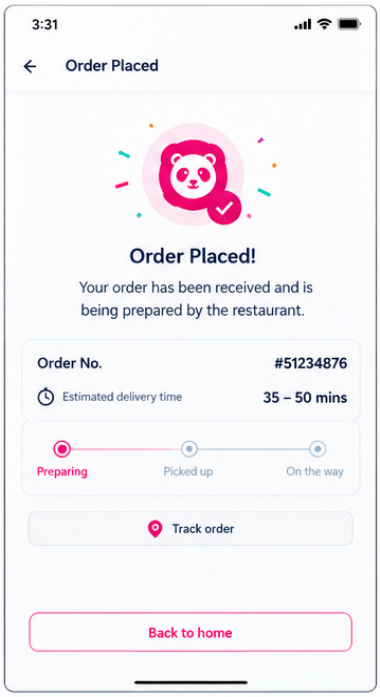
Principle violated: Error prevention and accuracy



3. Address form is hard to fill

Four required fields to type on a small screen, with no autofill or saved addresses. It is slow and easy to get wrong.

Principle violated: Flexibility and efficiency (substitutivity)



4. Unclear order confirmation

The screen shows an order number but not the restaurant or items, and the Track order button looks faint, like plain text.

Principle violated: Informative feedback and affordance

b. Scholarly article

Reference: Julia, M. B., & Rosell, J. (2024). Interaction and design barriers for older adults in food delivery apps: A usability study. *International Journal of Human Computer Interaction.* https://doi.org/10.1080/10447318.2024.2365484

Key finding. The researchers tested two food delivery apps with 12 older adults and found 22 and 23 interface barriers. Unclear feedback was among the most common causes of errors, since users could not tell what had happened after an action like adding a product to the cart. Most people finished the tasks, but not efficiently, and both apps scored below the System Usability Scale benchmark of 68.

Relation to my problems. This matches Problem 4, where the user is not fully sure what was ordered and where to check it. The study also found trouble telling buttons from plain text, which is the same issue as the faint Track order button. Adding the first product to the cart was among the slowest tasks too, which fits the tiny cart buttons in Problem 2. The participants were older adults, but the authors say fixing these barriers helps everyone.

The sketch below shows the improved checkout screen, confirmation screen and address form.

c. Design improvements

Problem	Improvement 1	Improvement 2
1. Wrong address suggestion	**Feedback and consistency.** Show the exact match first with its full address, and use "Did you mean" only when nothing matches.	**Error prevention.** Add a "Use my current location" button and a map pin so the user can confirm the address.
2. Tiny quantity buttons	**Affordance.** Make the plus and minus buttons large enough for a thumb, with clear spacing between them.	**Alignment and grouping.** Keep the controls in one aligned column on the right of each item, next to its price.
3. Address form	**Less typing.** Offer saved addresses (Home, Hostel) and autofill name and phone.	**Grouping and alignment.** Merge related fields, put labels above wide fields, show the right keyboard, and mark the rider note as optional.
4. Unclear confirmation	**Feedback.** Show restaurant name, items, total, order number and delivery time with a tick, plus a notification.	**Affordance and consistency.** Make Track order a filled primary button and keep Back to home as a lighter secondary button.

The sketch below shows the improved version of each screen.

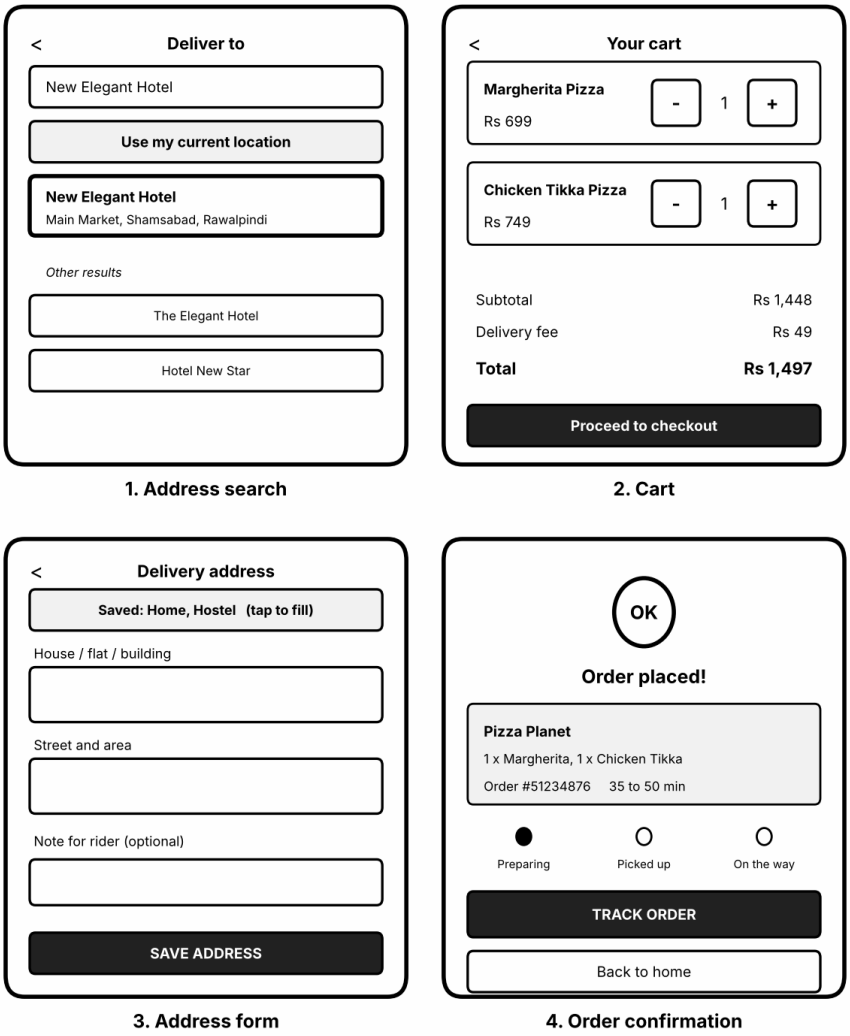


Figure 2: Sketch of the improved design

Question 3

a. Why skipping user research is a serious mistake

User focus means designing around real users and their needs, not what the designers imagine. Our course puts it simply: find out who the users are, remember they are probably not like you, talk to them and watch them. Skipping research breaks all of this.

EduConnect has three very different groups. A student on a phone between classes, a teacher entering marks for a whole class and an admin officer producing a report do not work the same way, and none of us has been in all three roles. The old portal shows what happens otherwise, with confusing navigation, slow marks upload and painful reports.

Time is a problem too. The system must be ready before next semester, so there is little room to fix big mistakes later, and problems found after launch cost far more than those found early. If users find it awkward they will avoid it, and the project wastes its money.

b. Personas

Persona 1: Student

Name and age	Sana Munir Alam, 20
Role	Third semester BS Computer Science student.
Background	Lives about 45 minutes away and travels by bus. Gives tuition two evenings a week, so her time is tight.
Technology use	Uses her phone for almost everything and opens her laptop only for assignments. Relies on mobile data, so heavy pages annoy her.
Goals	Check her schedule between classes, register before the deadline, see grades quickly and pay fees without standing in a queue.
Frustrations	The old portal crashes on registration day, its menus are confusing on a small screen, and she is never sure a payment went through.

Persona 2: Teacher

Name and age	Mr. Iqbal Hussain, 56.
Role	Senior lecturer in Computer Science, for 28 years.
Background	Teaches four courses with about 180 students. Works on the desktop in the staff room, prefers printed attendance sheets and keeps marks in Excel.
Technology use	Types slowly and is not comfortable with new software. Needs reading glasses, so small text and tiny buttons are a problem.
Goals	Upload marks accurately before the deadline, share slides with students and mark attendance without losing half the class time.
Frustrations	Marks upload is slow and he fears entering a wrong mark he cannot fix. EduConnect is the third new system he has to learn. He will try if it is simple and forgiving, but loses patience when the software makes him feel foolish.

c. Cultural probe methods

Method 1: Camera and screenshot log (students). For a week, a small group of students takes a photo or screenshot whenever they use or struggle with the old portal, with a short note on where they were and what went wrong. Students use it on buses and at home where a researcher cannot sit beside them, so this shows real situations like poor signal and gives material for interviews.

Method 2: Diary and postcard pack (teachers and admin staff). In a busy period like marks submission or report week, staff get a notebook with prompts such as "What did you try to do today?" and "How long did it take?", plus postcards to write one thing they wish the system could do. Teacher's dislike being watched while entering marks, so a diary lets them explain problems in their own words, and the postcards bring out ideas we would not think of.

d. Iterative design and prototyping

Requirements are never fully known at the start, and people often cannot say what they need until they see something. Iterative design builds a rough version, tests it, fixes what is wrong and repeats. With three user groups and a fixed deadline, this matters a lot for EduConnect. A mistake in a paper sketch takes minutes to fix, but in the finished system it can take weeks.

Two examples:

- **Admin report generation.** Give a few admin staff a prototype of the reports page and ask them to produce a fee defaulters list. It might show they cannot find the filters, or that they

expect an Export to Excel button. We rearrange the page before any real code is written, which answers the complaint about reports.

- **Course registration on mobile.** Let students try the registration steps on their own phones. It may show they cannot find Add Course or miss a timetable clash. We change the flow before launch, which answers the navigation complaint.

e. Wizard of Oz technique

In the Wizard of Oz technique the user thinks they are using a working system, but a person behind the scenes (the wizard) does the work the software would normally do.

For the EduConnect app, we could build only the screens and keep a team member in another room. When test student taps Pay Fee, the wizard changes the screen by hand to show a payment confirmation. To the student it feels like the real app.

It is useful because the backend and payment links do not need to exist yet. We can test the main student tasks early, see where people get confused and change the design cheaply, even trying costly ideas like a smart search box first. The wizard must respond quickly and consistently, or delays may look like design problems.

f. Shneiderman's Golden Rules

Rule	How to implement it	Example
1. Strive for consistency	Use one set of design standards across all three modules for layout, buttons, colours, wording and menus.	Save is always at the bottom right in the same colour, for marks, course choices and records alike.
3. Offer informative feedback	Respond to every action with a clear message, with progress indicators for slow tasks.	After a fee payment: "Payment of Rs 85,000 received" with a receipt number. After a marks upload: "Marks saved for 60 students".
5. Offer error prevention and simple error handling	Limit inputs to prevent mistakes, and explain any error in plain language with a fix.	The marks field accepts only 0 to the exam maximum, so 105 out of 100 is impossible. A course clash names the two overlapping courses and suggests alternatives.
8. Reduce short term memory load	Keep key information on screen and let users pick from lists instead of remembering.	A registration summary panel keeps chosen courses and credit hours visible, and the admin report page offers recent settings in a dropdown.

g. Hierarchical diagram

After login, each group goes to its own module, and every task is one step below it.

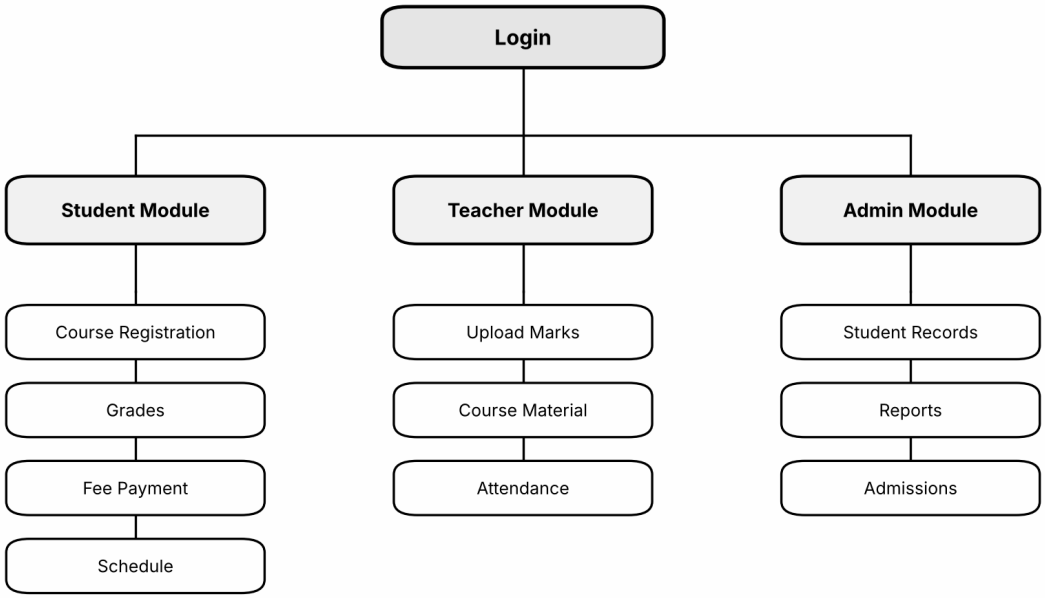


Figure 3: Hierarchical structure of EduConnect